



PROJECT REPORT

**Project title : TongaKids Read Phonics Engine
for Chitonga-Speaking Learners**

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Table of Contents

ABSTRACT.....	3
CHAPTER 1: INTRODUCTION.....	4
1.1 INTRODUCTION.....	4
1.2 BACKGROUND.....	4
1.3 PROBLEM STATEMENT.....	5
1.4 OBJECTIVES.....	6
Main Objective:.....	6
1.5 SCOPE, PURPOSE AND APPLICABILITY.....	7
CHAPTER 2: LITERATURE REVIEW.....	8
2.1 INTRODUCTION.....	8
2.2 RELATED WORK.....	8
2.3 PREVIOUS SYSTEMS OR RELATED APPLICATIONS.....	9
1. LubutoLiteracy (Zambia):.....	9
2. GraphoGame Zambia (RESUZ Project):.....	10
3. Tau Laukonga (Pacific Region.....	11
2. The Core Contradiction.....	12
2.4 CONCLUSION.....	12
CHAPTER 3: RESEARCH METHODOLOGY.....	14
3.1 INTRODUCTION.....	14
3.2 METHODOLOGY.....	15
3.3 INFORMATION GATHERING AND ANALYSIS.....	15
3.3.1 INFORMATION GATHERING.....	15
3.3.1 TECHNICAL REQUIREMENTS.....	16
3.3.2CONCLUSION.....	16
3.4 REQUIREMENTS SPECIFICATION.....	16
3.4.1 INTRODUCTION.....	16
3.4.2 FUNCTIONAL REQUIREMENTS.....	16
3.4.3 NON-FUNCTIONAL REQUIREMENTS.....	16
3.4.4 ASSUMPTIONS AND CONSTRAINTS.....	17
3.4.5 CONCLUSION.....	17

CHAPTER 4: SYSTEM DESIGN.....	18
4.1 INTRODUCTION.....	18
3-TIER ARCHITECTURE.....	19
USE CASE DIAGRAMS.....	21
CLASS DIAGRAM.....	26
SEQUENCE DIAGRAM.....	27
ACTIVITY DIAGRAM.....	28
STATE DIAGRAM.....	29
References.....	30

TABLE OF FIGURES

Figure 1lubuto website.....	9
Figure 22.GraphoGame Zambia (RESUZ Project):.....	10
Figure 33Tau Laukonga.....	12
Figure 4 3-TIER ARCHITECTURE.....	21
Figure 5USE CASE DIAGRAMS.....	22
Figure 6CLASS DIAGRAM.....	27
Figure 7SEQUENCE DIAGRAM.....	28
Figure 8 ACTIVITY DIAGRAM.....	29
Figure 9STATE DIAGRAM.....	30

ABSTRACT

TongaKids Read Phonics Engine for Chitonga-Speaking Learners

focuses on solving the foundational learning crisis in Zambia where a vast majority of children struggle to understand age-appropriate text. While the government has mandated mother-tongue instruction, the transition has been hindered by a critical lack of supplementary materials, especially in the Southern Province where the book-to-pupil ratio remains stagnant at 1:7. The problem is worsened by existing digital tools that use foreign imagery and Germanic phonetic rules such as references to "snow" which are entirely alien to rural learners and result in poor engagement.

The proposed system is a mobile-native Android application designed to provide a culturally resonant pedagogical bridge for Grade 1 to 4 learners. The primary objectives are to develop a platform that teaches basic Chitonga phonics and reading skills using fifty illustrated folktales and synchronized audio narration. To ensure usability in districts like Choma and Gwembe, the system is built with an offline-first architecture to function without constant internet access. The app also includes self-paced progress tracking to support individualized learning in overcrowded classrooms where teacher attention is often limited.

The project follows the Agile Software Development Life Cycle (SDLC) combined with Object-Oriented Analysis and Development (OOAD), allowing for flexible, iterative improvements based on stakeholder feedback. The research is based on the hypothesis that a phonics engine tailored specifically to the transparent orthography of Chitonga will significantly improve letter-sound identification and oral reading fluency. By leveraging the high rate of mobile phone ownership, the study aims to demonstrate a scalable and sustainable technical solution for foundational literacy in low-resource environments.

CHAPTER 1: INTRODUCTION

1.1 INTRODUCTION

Foundational literacy is globally recognized as the primary predictor of a child's long term economic prosperity and academic achievement. Despite its critical importance, the "Foundational Learning Crisis" has emerged as a paramount concern for technologists and policymakers across Sub-Saharan Africa. In the Zambian context, recent national assessments indicate a staggering crisis where approximately 98.5 to 99 percent of children at age ten are unable to read or understand a simple age-appropriate text. This high rate of "Learning Poverty" in Zambia is 13 percentage points higher than the average for the Sub-Saharan Africa region, directly threatening the nation's ability to build sustainable human capital. Early intervention is identified as the most viable solution to this gap, as studies show that interest in literacy can be successfully sparked from a young age through engaging activities that utilize transparent linguistic structures. Consequently, this research project proposes the "TongaKids Read" system, a mobile-native application designed to introduce primary school children to foundational phonics at an early age through interactive storytelling. The application recognizes the urgent need for inclusive, accessible, and high-quality digital learning resources that can be reached by all learners regardless of their location.

1.2 BACKGROUND

The field of education in Zambia has undergone a significant transformation characterized by a shift from English-only instruction back to a mandated mother-tongue approach for early grades. Historically, the "Straight for English" policy, rooted in the colonial administrative legacy, created systemic barriers for millions of learners in rural provinces who entered the formal school system with no exposure to the English language. In response to these persistent failures, the Ministry of Education enacted the 2014 Language in Education policy, mandating that initial literacy instruction from Grade 1 to Grade 4 be conducted in one of the seven officially recognized regional languages, including Chitonga. Chitonga is an ideal candidate for such intervention because it possesses a transparent orthography, where written symbols correspond directly and consistently to specific speech sounds. Furthermore, rural learners often face a "dialect mismatch" between the standard language used in textbooks and the colloquial forms spoken in their communities. As traditional print infrastructure remains underdeveloped, Zambia has witnessed a "leapfrogging" in technology where mobile telecommunications have expanded rapidly. Educational games have the potential to make learning more engaging and accessible for young female and male learners in this environment. By incorporating local folktales and cultural references, developers can create content that resonates with the Zambian child and fosters a sense of pride and identity. This project aligns with the National Education Policy 2025 and the Zambia Education Curriculum Framework 2023 by exploring the effectiveness of game-based learning in delivering foundational literacy skills.

1.3 PROBLEM STATEMENT

The central problem addressed by this study is the acute shortage of linguistically appropriate and culturally resonant digital literacy tools for Chitonga-speaking children in Zambia's Southern Province. While the government's 2014 policy identified the mother tongue as the optimal medium for initial literacy, the transition from policy to practice has been hindered by a critical lack of supplementary materials. Most existing digital educational products in the Zambian market are re-skinned versions of Western software featuring phonics rules and imagery, such as snow or European wildlife, that are entirely alien to a child in a rural village. This "scarcity of relevance" results in low engagement and poor learning outcomes, as evidenced by assessments showing that 65 percent of Grade 2 learners could not read a single word. The problem is further compounded by the linguistic transition gap at Grade 5, where students struggle to master the orthographically opaque rules of English without a solid foundation in the transparent orthography of their mother tongue. This failure often leads to high dropout rates and poor performance in the Grade 7 Composite Examination, which determines a child's access to secondary education. Furthermore, technical barriers such as intermittent power supply and high data costs render many cloud-based platforms useless in districts like Choma and Chibombo. Our project seeks to address this problem by creating an inclusive, mobile-native, and offline-capable learning environment that utilizes localized design principles to improve foundational literacy.

1.4 OBJECTIVES

Main Objective:

To design, develop, and evaluate the TongaKids Read mobile application to improve literacy among Chitonga-speaking learners in Grades 1–4 in Zambia.

Specific Objectives

To develop a mobile application that teaches basic Chitonga phonics and reading skills.

To include interactive learning content such as stories, sounds, and reading exercises.

To ensure the application works offline for use in rural areas without internet access.

To design a simple and child-friendly user interface suitable for early-grade learners.

To allow audio narration to support pronunciation and reading comprehension and track learner progress within the application for self-paced learning.

1.5 SCOPE, PURPOSE AND APPLICABILITY

This study is geographically and demographically focused on the Southern Province of Zambia, specifically the districts of Choma, Monze, and Chibombo, which represent a mix of rural and semi urban communities where literacy levels remain low. Technically, the document covers the full Agile Software Development Life Cycle (SDLC) of the TongaKids Read application for the Android platform, which is the most commonly used mobile operating system in Zambia. The research utilizes an iterative approach to minimize risks such as bugs and changing requirements while ensuring the end product is robust and adaptable. For the system to be effective, it is assumed that users have occasional access to a mobile device and an initial internet connection to download the app. Two major limitations arise: the digital divide regarding hardware access and high data costs which may limit the reach of the intervention among the poorest households.

2.1 INTRODUCTION

The literature review serves as the critical foundation for the TongaKids Read project, offering a systematic examination of existing research and the development of educational tools aimed at improving early-grade literacy through mother-tongue instruction. This review provides a contextual overview of the "Foundational Learning Crisis" in Zambia, where recent national assessments indicate that approximately 98.5 to 99 percent of ten-year-old children are unable to read or understand a simple age-appropriate text (World Bank, 2019; World Bank, 2022). This staggering rate of "Learning Poverty" is 12.2 percentage points higher than the average for the Sub-Saharan Africa region, underscoring the urgent need for evidence-based interventions (World Bank, 2022). The purpose of this review is to explore how the intersection of mother-tongue instruction, Information Communication Technology (ICT), and game-based learning can address these systemic failures (Hanzooma, K 2024). Specifically, it examines the barriers to literacy acquisition in the Southern Province, where the current student-to-textbook ratio remains at an inadequate 1:7 (Ministry of Education, 2014). By synthesizing global and regional efforts in digital literacy, this review identifies the "Literature Gap" that TongaKids Read aims to fill: the need for a fully integrated, mobile-native solution that combines Chitonga-specific phonics with culturally resonant storytelling (Pearlson, K. E, 2024).

2.2 RELATED WORK

The evolution of educational technology in Sub-Saharan Africa is characterized by a "leapfrogging" phenomenon, where traditional print infrastructure is bypassed in favor of mobile telecommunications (Nishanthi, R. 2020). Research indicates that while traditional classroom instruction often focuses on whole-language approaches, phonics-based instruction is specifically identified as the most effective method for developing reading skills in resource-constrained environments (RTI International, 2016). Scientific evidence highlights that literacy acquisition is heavily influenced by "orthographic depth"; Chitonga possesses a "transparent orthography" where written symbols correspond directly to individual speech sounds, making it significantly easier for beginning readers to master compared to the "opaque" rules of English (Kaani et al., 2021;). However, many digital tools deployed in Africa fail to leverage this advantage because they are re-skinned versions of Western software featuring imagery such as snow or European wildlife that are entirely alien to rural Zambian learners (World Bank, 2019;). This scarcity of relevance leads to a cultural disconnect that hinders cognitive development and engagement (Olubendi, D. 2019). Educational games provide a solution to these cultural and pedagogical barriers by offering a fun, interactive medium that fosters a sense of pride and identity through localized narratives .

2.3 PREVIOUS SYSTEMS OR RELATED APPLICATIONS

Several initiatives have attempted to address foundational literacy through digital means, providing direct design inspiration for the TongaKids Read engine:

1. **LubutoLiteracy (Zambia):** This project developed high-quality mother-tongue learning content across seven Zambian languages, including Chitonga (Lubuto Library Partners, 2024). Originally programmed using the Etoys application for durably-built One Laptop Per Child (OLPC) hardware, the initiative reached over 10,000 children (eIFL.net, 2014). However, evaluations revealed that the project's reliance on specific hardware and its lack of automated feedback mechanisms limited its reach and pedagogical efficiency in rural clusters (IFLA, 2024; eIFL.net, 2014)

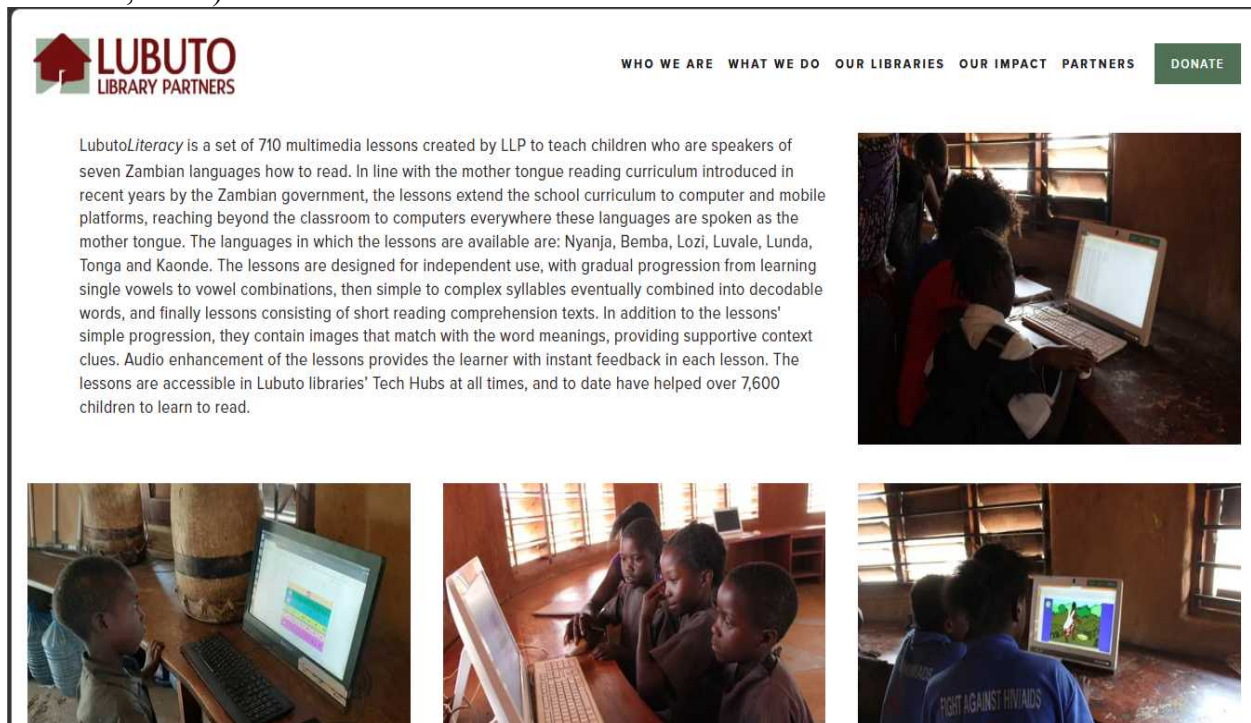


Figure 1lubuto website

2. **GraphoGame Zambia (RESUZ Project):** Developed as an adaptation of a Finnish dyslexia prevention game, GraphoGame focuses on improving letter-sound recognition and decoding skills through mobile phones (Ahmed et al., 2020). Intervention studies in the Lusaka district showed significant improvements in participants' decoding skills (Tale Africa, 2021). Despite these technical successes, the system was limited by a lack of cultural storytelling, which researchers found reduced long-term reading motivation compared to narrative-rich interventions).



Figure 22. GraphoGame Zambia (RESUZ Project):

3. **Tau Laukonga (Pacific Region):** This mobile application, translated as "Let's Read," provides a force multiplier for literacy by offering a free, offline library of over 100 culturally relevant books (Friends of Tonga, 2025; MWM AI, 2025). The app features native audio narration to help children master correct accents and tones, bridging the gap between home languages and formal instruction (MWM AI, 2025). Its success in functioning without a consistent internet connection provides a technical blueprint for the "offline-first" architecture of TongaKids Read (Friends of Tonga, 2025).



Figure 33Tau Laukonga

The following table compares the established systems against the design goals of TongaKids Read:

table comparison of the established systems against the design goals

System	Primary Strength	Critical Weakness/Critique
GraphoGame Zambia	Phonics Accuracy: Highly effective at improving letter-sound recognition and basic decoding skills through structured drills.	Engagement Deficit: Lacks a narrative framework. Research indicates that its absence of storytelling and cultural content reduces long-term reading motivation.
Lubuto Digital Library	Content Depth: Provides extensive mother-tongue learning materials across multiple Zambian languages.	Hardware Dependency: Originally built for laptop-based systems (OLPC), which reduces accessibility in rural clusters where smartphones are the primary digital entry point.
TaRL Digital Tools	Pedagogical Pace: Uses adaptive learning principles that allow learners to progress based on their actual level rather than their grade.	Resource Requirements: Many adaptive systems require consistent data synchronization and teacher intervention, which is difficult to maintain in low-resource environments.

2. The Core Contradiction

The fundamental contradiction that **TongaKids Read** resolves is the "**Localization vs. Technical Reach**" gap.

- **The Problem** Existing digital tools in the Zambian market are frequently adaptations of Western software. These systems create a pedagogical contradiction by using "transparent" Chitonga phonics but "opaque" Western imagery that are entirely alien to rural Zambian learners.
- **The Technical Gap:** While initiatives like the Lubuto Digital Library are culturally rich, their reliance on specific hardware creates a barrier to scale. Conversely, mobile-native tools like GraphoGame are technically accessible but culturally Hardware Inclusivity: It bypasses the laptop-dependency of the Lubuto system by optimizing for the entry-level Android devices already owned by parents in Southern Province.

2.4 CONCLUSION

In conclusion, the literature confirms that game-based learning offers significant opportunities to increase interest in foundational literacy by making instruction fun, interactive, and mobile. The review synthesized scientific evidence regarding the effectiveness of phonics-based methods in transparent Bantu orthographies and

highlighted the role of cultural relevance in sustaining learner motivation (Kaani et al., 2021; RTI International, 2016). However, the study identifies a persistent Literature Gap: existing systems in Zambia are often hardware-dependent, lacking in cultural narrative, or technically limited by low-bandwidth communication channels (IFLA, 2024; Tale Africa, 2021). Furthermore, the "digital divide" and high data costs remain significant technical challenges that hinder the acceptance of cloud-dependent tools in rural districts like Choma and Gwembe (World Bank, 2019). The TongaKids Read project addresses these challenges by engineering an offline-first phonics engine that integrates fifty illustrated folktales with synchronized audio narration. This integrated approach provides a scalable and sustainable technical solution to the foundational learning crisis by leveraging the high rate of mobile phone ownership in Zambia.

3.1 INTRODUCTION

The system development methodology used to develop the TongaKids Read project is the Agile method (Al-Saqqa et al., 2020). The rationale for selecting Agile methodology is its iterative and incremental approach, which focuses on delivering functional software in short, manageable periods (Da Silva, 2010; MDPI, 2021). Unlike the rigid waterfall methodology, Agile allows for continuous changes throughout the development process, making it significantly more flexible and responsive to the evolving requirements of a localized literacy intervention (Eason, 2016; Pearlson et al., 2024). Below are the procedures, techniques, and tools used to investigate the implementation of this system for primary school learners (World Bank, 2019).

- **Theoretical Framework:** This study is anchored on the assumption that game-based learning is an effective method for changing perspectives on subjects often perceived as difficult, such as literacy and STEM-related skills (Mulundu et al., 2021; RTI International, 2016). By acquired foundational skills in a familiar language, the system aims to increase participation and confidence among learners in marginalized communities (Kioko, 2015; Nishanthi, 2020).
- **Research Design:** The research utilizes a quantitative research design to evaluate the effectiveness of educational games in introducing concepts of phonics and decoding (Ahmed et al., 2020). The design includes a pre-test and post-test study to compare the knowledge and proficiency levels of participants before and after playing the game, assessing the system's efficiency relative to traditional teaching methods (Kaani et al., 2021; Tale Africa, 2021).
- **Data Collection:** Data will be collected through structured interviews and systematic observation of system usage (Olubendi, 2019). Interviews include questions regarding the participants' user experience, satisfaction levels, and their evolving perception of literacy (UNICEF, 2016). Observation focuses on the "time-on-task"—measuring the duration taken to complete a level—and the overall user experience during gameplay (Mulundu et al., 2021).
- **Ethics Considerations:**
 - a. In accordance with the Zambia Data Protection Act 2021, explicit informed consent is obtained from guardians and school administrators before participation (Zambia Data Protection Act, 2021).
 - b. Participants' privacy is maintained through pseudonymization, utilizing de-identified interview transcripts to protect identity (Winschiers-Theophilus & Bidwell, 2013).

c. The study strictly adheres to international ethical guidelines for research involving human subjects and children (UNICEF, 2017).

3.2 METHODOLOGY

A Software Development Life Cycle (SDLC) is a framework used for designing, developing, and testing high-quality software to meet customer requirements within defined constraints (Somerville, 2015). This project utilizes the Agile SDLC combined with an Object-Oriented Analysis and Development (OOAD) approach to define system architecture and data flow (Beck et al., 2001). Agile SDLC breaks down the product into small incremental builds delivered in iterations (Zia, 2018). The phases are as follows:

I. Requirement Gathering and Analysis: System requirements are determined through brainstorming sessions and categorization based on priority (Kumar et al., 2019).

II. Design the Requirements: The development team translates specific requirements into technical designs, ensuring the phonics engine adheres to Chitonga orthography (Kaani et al., 2021).

III. Construction/Iteration: Features are implemented in multiple iterations, with revisions based on deliverables at the end of each cycle (Mindster, 2024).

IV. Deployment: The working system is deployed in a controlled pilot environment for evaluation (World Bank, 2019).

V. Testing: Selected individuals test basic functionality to ensure the system gives expected results with specific inputs (MDPI, 2021).

VI. Feedback: The final product is delivered to stakeholders, such as the school of Information and Communication Technology, for review (Al-Saqqa et al., 2020).

The Agile methodology ensures flexibility and continuous improvement through 2–4 week sprints (Bhavsar et al., 2020). Weekly stand-up meetings promote transparency and collaboration within the project team (Kaleshovska et al., 2015). By adopting this approach, the project aims to deliver a high-quality application that effectively teaches phonics to primary school girls and boys in an interactive way; Winschiers-Theophilus & Bidwell, 2013).

3.3 INFORMATION GATHERING AND ANALYSIS

The purpose of this gathering process is to identify user needs for an educational game designed to introduce children to foundational literacy (World Bank, 2019).

3.3.1 INFORMATION GATHERING

1. **Surveys:** Existing surveys were used to collect feedback on desired mobile application features (Mulundu et al., 2021).
2. **Research:** The team researched best practices for "Africanized" HCI and the psychology of game-based learning (Winschiers-Theophilus & Bidwell, 2013).

3.3.1 TECHNICAL REQUIREMENTS

- **Mobile-Based Application:** The system must run on mobile devices, as these are the most accessible technology for Zambians (World Bank, 2024).
- **User Interface:** Designs must be age-appropriate and child-friendly (Winschiers-Theophilus & Bidwell, 2013).
- **Security:** Encryption protects sensitive data transmitted between the user and the system (World Bank, 2019).
- **Scalability:** The application must accommodate an increasing number of users without performance degradation (Al-Saqqa et al., 2020).

3.3.2 CONCLUSION

By gathering and analyzing stakeholder information, the team identified technical requirements that inform the implementation of the educational game, ensuring it maintains the integrity of regional literacy standards (Ministry of Education, 2023).

3.4 REQUIREMENTS SPECIFICATION

3.4.1 INTRODUCTION

The system is a 2D mobile-based game designed with the Unity engine to address the foundational learning crisis in Zambia (World Bank, 2019; Ministry of Education, 2023).

3.4.2 FUNCTIONAL REQUIREMENTS

- **User Registration:** Users can register using an email address or phone number (Zambia Data Protection Act, 2021).
- **Authentication:** Supports secure login mechanisms including password recovery (World Bank, 2019).
- **User Profile Management:** Users can customize profiles and usernames (Mulundu et al., 2021).
- **Progress Tracking:** Tracks success in phonics lessons using badges, stars, and leaderboards to motivate the child (Ahmed et al., 2020).
- **Educational Lessons:** Includes interactive coding-inspired phonics lessons tailored for primary school learners (Kaani et al., 2021).

3.4.3 NON-FUNCTIONAL REQUIREMENTS

- **Usability:** Clear instructions and intuitive navigation for emergent readers (Winschiers-Theophilus & Bidwell, 2013).
- **Performance:** The system must load quickly and maintain smooth performance on low-specification devices (World Bank, 2024).
- **Reliability:** Continuous operation with minimal downtime and robust error handling (Somerville, 2015).
- **Scalability:** Ability to support a growing user base across the Southern Province (Al-Saqqa et al., 2020).

- **Security:** Adherence to data sovereignty requirements, hosting children's data locally in Zambia (Zambia Data Protection Act, 2021).

3.4.4 ASSUMPTIONS AND CONSTRAINTS

- **Hardware Requirements:** The system requires a mobile device with a minimum screen size (World Bank, 2024).
- **Connectivity:** Internet connection is required for the initial download of the application (World Bank, 2019).
- **Accessibility:** Accessible to all eligible users regardless of their financial background (SDG 4; UNESCO, 2013).
- **Legal Compliance:** Compliance with Zambian children's rights and media laws (Zambia Data Protection Act, 2021).

3.4.5 CONCLUSION

By incorporating these functional and non-functional requirements, the system ensures an interactive learning experience while maintaining the security and privacy of users (Ministry of Education, 2023).

CHAPTER 4: SYSTEM DESIGN

4.1 INTRODUCTION

The design phase of the TongaKids Read project represents the critical transition from understanding the foundational learning crisis in Zambia to engineering a robust, technical solution capable of operating within the socio-technical constraints of the Southern Province. As established in the preceding analysis of the Zambian educational landscape, the move from an English-only instruction policy to a structured mother-tongue approach has created a scarcity of relevance in pedagogical materials. Consequently, the system design detailed herein is not merely a blueprint for a mobile application, but a comprehensive framework for a digital literacy intervention that integrates rule-based phonics, culturally resonant storytelling, and a resilient, offline-first data architecture.

The engineering process for TongaKids Read is governed by the principles of Object-Oriented Analysis and Development (OOAD), which allows for the decomposition of complex educational tasks into manageable, interactive components.

3-TIER ARCHITECTURE

For the TongaKids Read project, a 3-Tier Architecture is the most effective way to ensure the application remains robust, secure, and most importantly functional without an internet connection. This architecture divides the system into three logical layers that communicate through clearly defined interfaces.

1. The Presentation Tier (The Front-End)

This is the top-most level of the application. It is the only part of the system the Learner and Parent see directly.

- Platform: Android Mobile Application.
- Responsibility: Handling user interactions, rendering the 50 illustrated folktales, and playing the synchronized audio clips.
- Key Feature: The "Parental Gateway" is a distinct sub-module within this tier that requires a security challenge (PIN) before allowing access to the data layer.

2. The Application Tier (The Logic Layer)

This is the brain of the system. It processes the inputs from the Presentation Tier and decides what data is needed from the bottom tier.

- Location: Resides locally on the Android device's CPU.
- Core Engines:
 - Phonics Engine: Converts user taps into requests for specific Chitonga phonemes.
 - Progress Calculator: Analyzes quiz performance to determine "Reading Accuracy" and "Words Per Minute."
 - Data Controller: Implements the Offline-First logic, ensuring that if no network is detected, it defaults strictly to local operations.

3. The Data Tier (The Storage Layer)

This tier is responsible for the persistence of all information. In this project, it is split between the local device and a remote server.

- Local Component (SQLite/Room): Stores the actual text of the folktales, audio file paths, and the learner's historical metrics. This allows for 100% offline functionality.
- Remote Component (PostgreSQL/Cloud): Acts as a backup. When the device connects to Wi-Fi, the Application Tier triggers a "Sync" to move pseudonymized data to the cloud for regional education reporting.
- Security: This layer ensures that learner data is encrypted, adhering to the Zambia Data Protection Act.

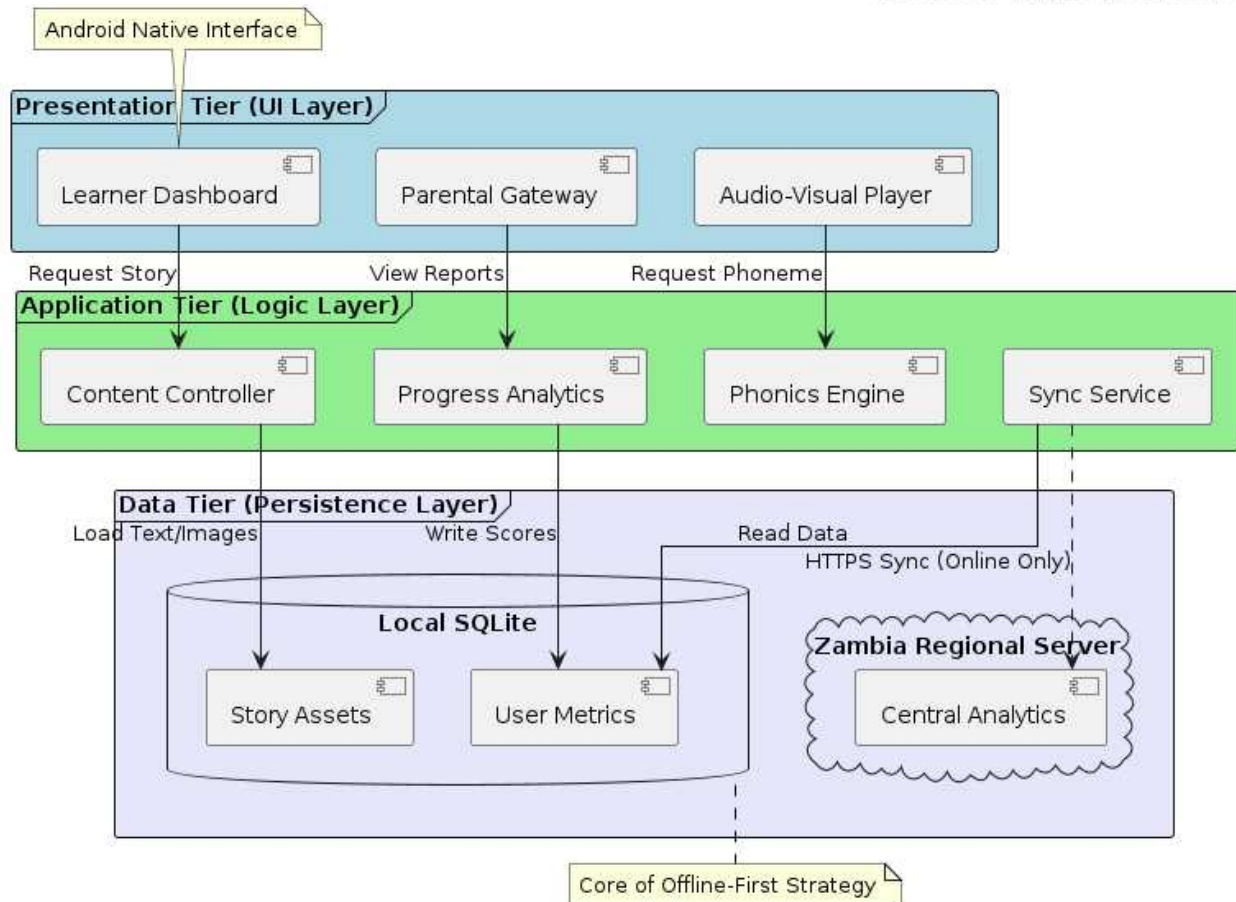


Figure 4 3-TIER ARCHITECTURE

USE CASE DIAGRAMS

The Use Case Diagram serves as the functional blueprint for the TongaKids Read application. It illustrates the boundaries of the system and how two distinct actors the Learner and the Parent/Guardian interact with its core features. By mapping out these relationships, the diagram ensures that both pedagogical needs (literacy) and legal requirements (data consent) are addressed within the software architecture.

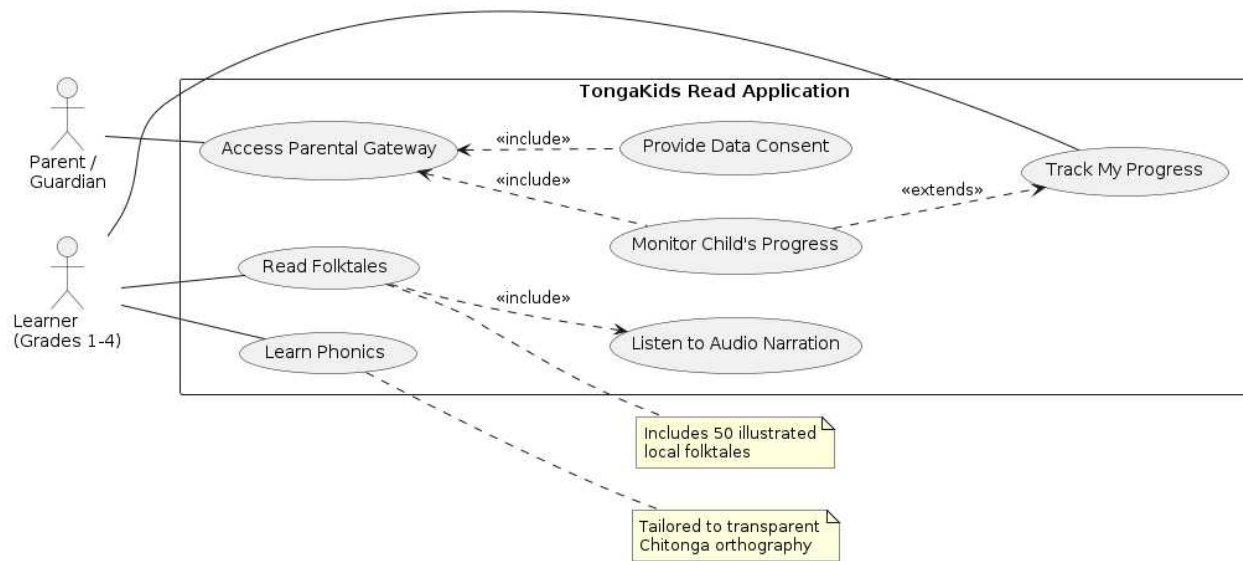


Figure 5 USE CASE DIAGRAMS

Use case descriptions for the TongaKids Read system,

1. Use Case: Learn Phonics (UC1)

This use case covers the core instructional engine where students interact with the unique sounds of the Chitonga language.

Field	Details
Use Case ID	UC-01
Use Case Name	Learn Phonics
Primary Actor	Learner (Grades 1-4)
Description	The learner interacts with phonemic modules to master letter-sound identification and syllable blending in Chitonga.
Preconditions	The application is open and a Learner profile is active.
Main Success Scenario	1. Learner selects a specific phonics level from the menu. 2. System displays syllables based on transparent orthography. 3. Learner taps a syllable to trigger the native audio pronunciation. 4. Learner completes a matching game or quiz. System provides immediate visual/audio feedback
Postconditions	Accuracy data is updated in the local progress tracker

2. Use Case: Read Illustrated Folktales (UC2)

This use case describes the primary content delivery method, integrating local culture with literacy training.

Field	Details
Use Case ID	UC-02
Use Case Name	Read Illustrated Folktales
Primary Actor	Learner (Grades 1-4)

Description	The learner browses and reads a library of 50 localized stories with supportive multimedia.
Preconditions	At least one story package is downloaded/available offline.
Main Success Scenario	1. Learner opens the Digital Library. 2. Learner selects a story (e.g., a traditional Zambian fable). 3. System renders high-quality illustrations and text. 4. System activates synchronized audio narration (Include: UC3). 5. Learner navigates through the pages.
Extensions	UC3 (Listen to Audio): Triggered automatically or manually to support multimodal learning.
Postconditions	The story is marked as "Read" in the user's history.

3. Use Case: Access Parental Gateway (UC5)

This use case represents the administrative and legal layer of the application for guardians.

Field	Details
Use Case ID	UC-05
Use Case Name	Access Parental Gateway
Primary Actor	Parent / Guardian
Description	A secure area where parents manage settings, legal consent, and view reports.
Preconditions	The app is in a state allowing navigation to settings.
Main Success Scenario	1. Parent selects the "Parental Gateway" icon.

	2. System presents a security challenge (e.g., a simple math problem or PIN). 3. Parent successfully authenticates. 4. System provides access to Consent (Include: UC6) and Monitoring (Include: UC7).
Postconditions	Parent gains access to sensitive data and configuration controls.

4. Use Case: Monitor Learner Progress (UC7)

This describes how the system visualizes the tracked data for the guardian to evaluate performance.

Field	Details
Use Case ID	UC-07
Use Case Name	Monitor Learner Progress
Primary Actor	Parent / Guardian
Description	The parent views analytical data regarding the child's reading speed and accuracy.
Preconditions	Parent is logged into the Parental Gateway;

	child has completed at least one session.
Main Success Scenario	1. Parent selects "Progress Dashboard". 2. System retrieves data from the device's local database (Extend: UC4). 3. System displays a summary of words read per minute and phonics scores. 4. Parent identifies specific areas where the child is struggling.
Postconditions	The parent is informed of the learner's foundational literacy status.

CLASS DIAGRAM

The Class Diagram maps the relationships between these entities, showing how they inherit properties from one another or work together to achieve the system's goals. For this project, the diagram specifically highlights how the application manages "Offline-First" data storage and the unique phonetic structures of the Chitonga language.

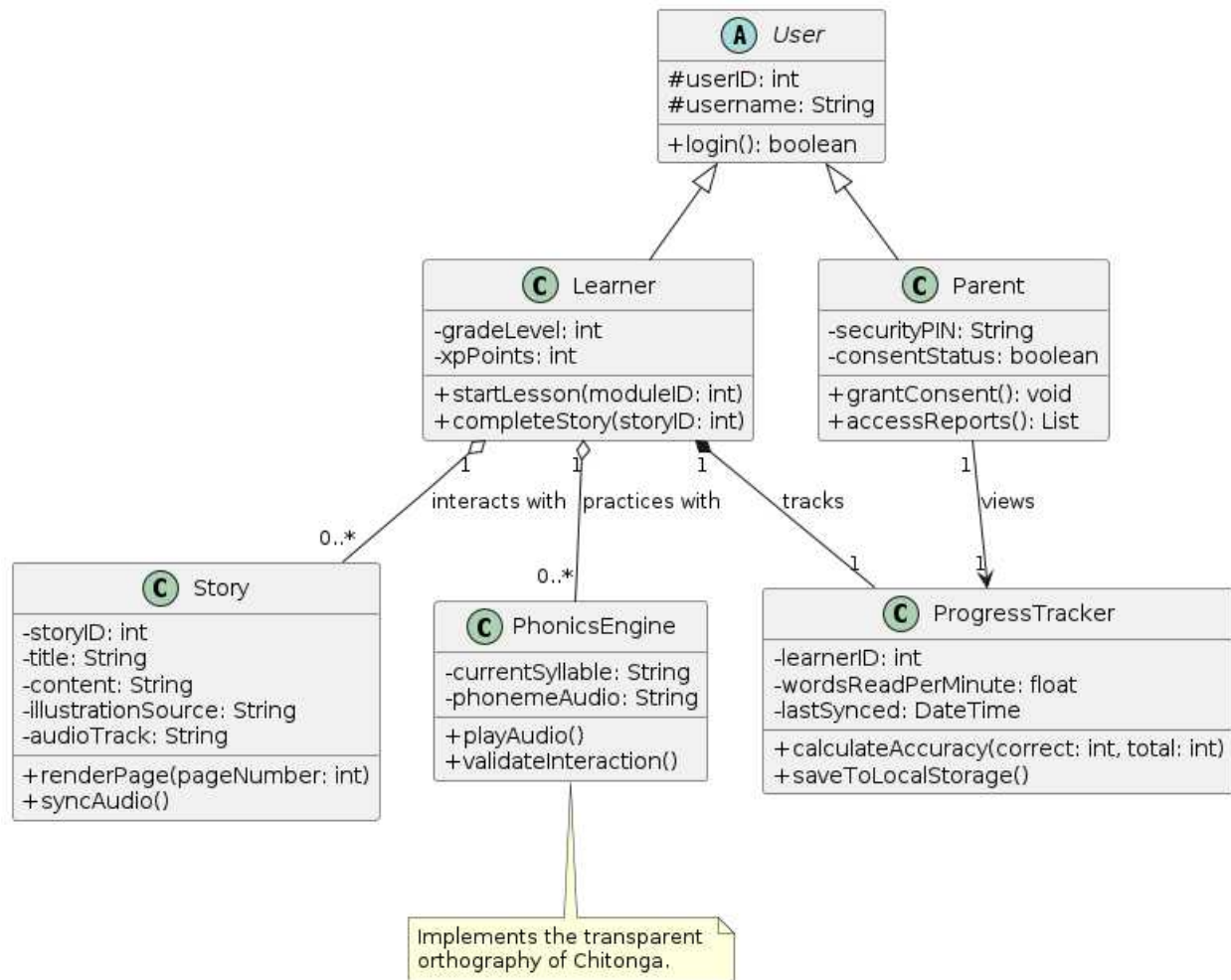


Figure 6CLASS DIAGRAM

SEQUENCE DIAGRAM

The Sequence Diagram captures the dynamic behavior of the TongaKids Read system, focusing on how different software components collaborate over time to complete a specific task. Unlike static diagrams that show what the system *is*, the sequence diagram shows how the system *functions* during an active user session.

This specific diagram models two critical workflows: the Phonics Practice Session and the Progress Logging phase. It illustrates the real-time exchange of messages between the Learner, the User Interface, and the Back-end Engines, highlighting the system's ability to provide immediate multimodal feedback while securely managing learner data according to Zambian data protection standards.

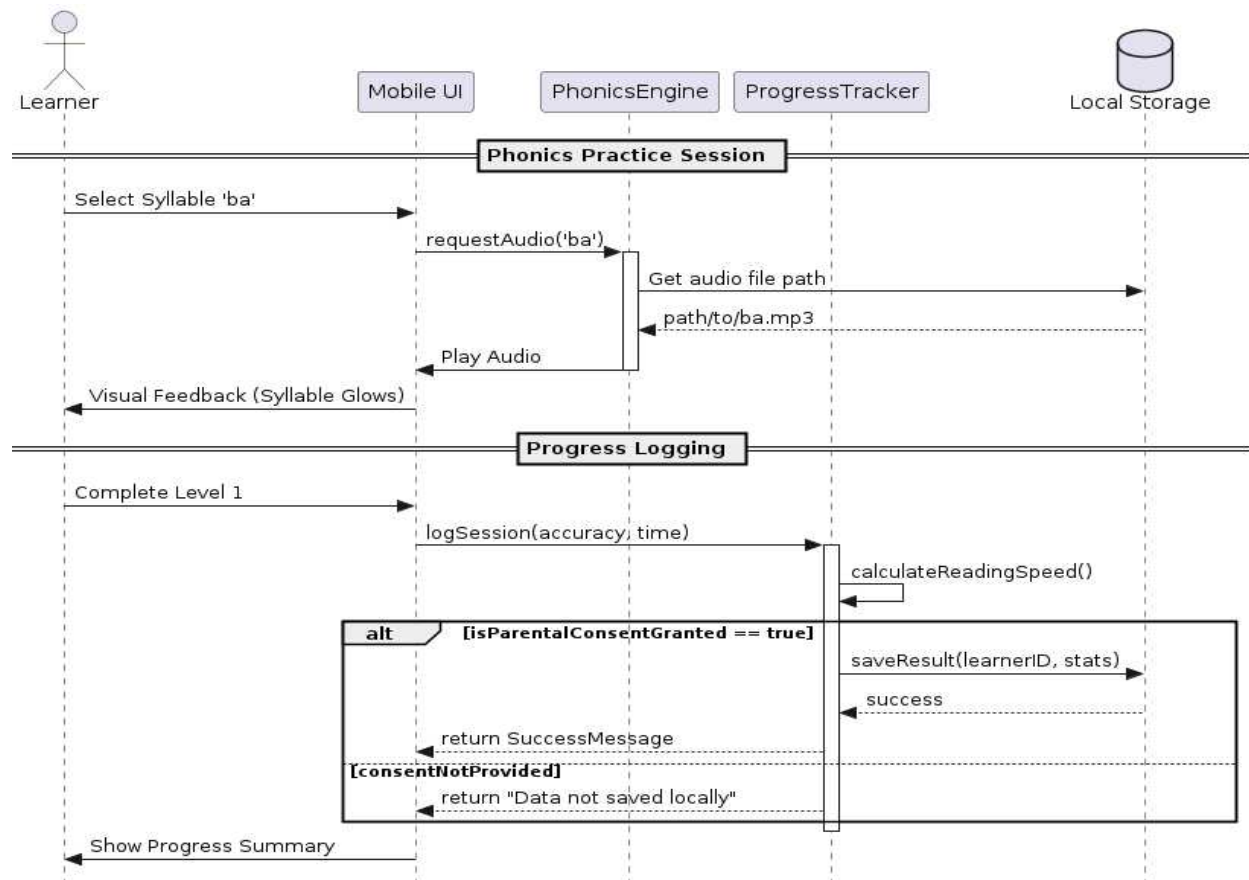


Figure 7SEQUENCE DIAGRAM

ACTIVITY DIAGRAM

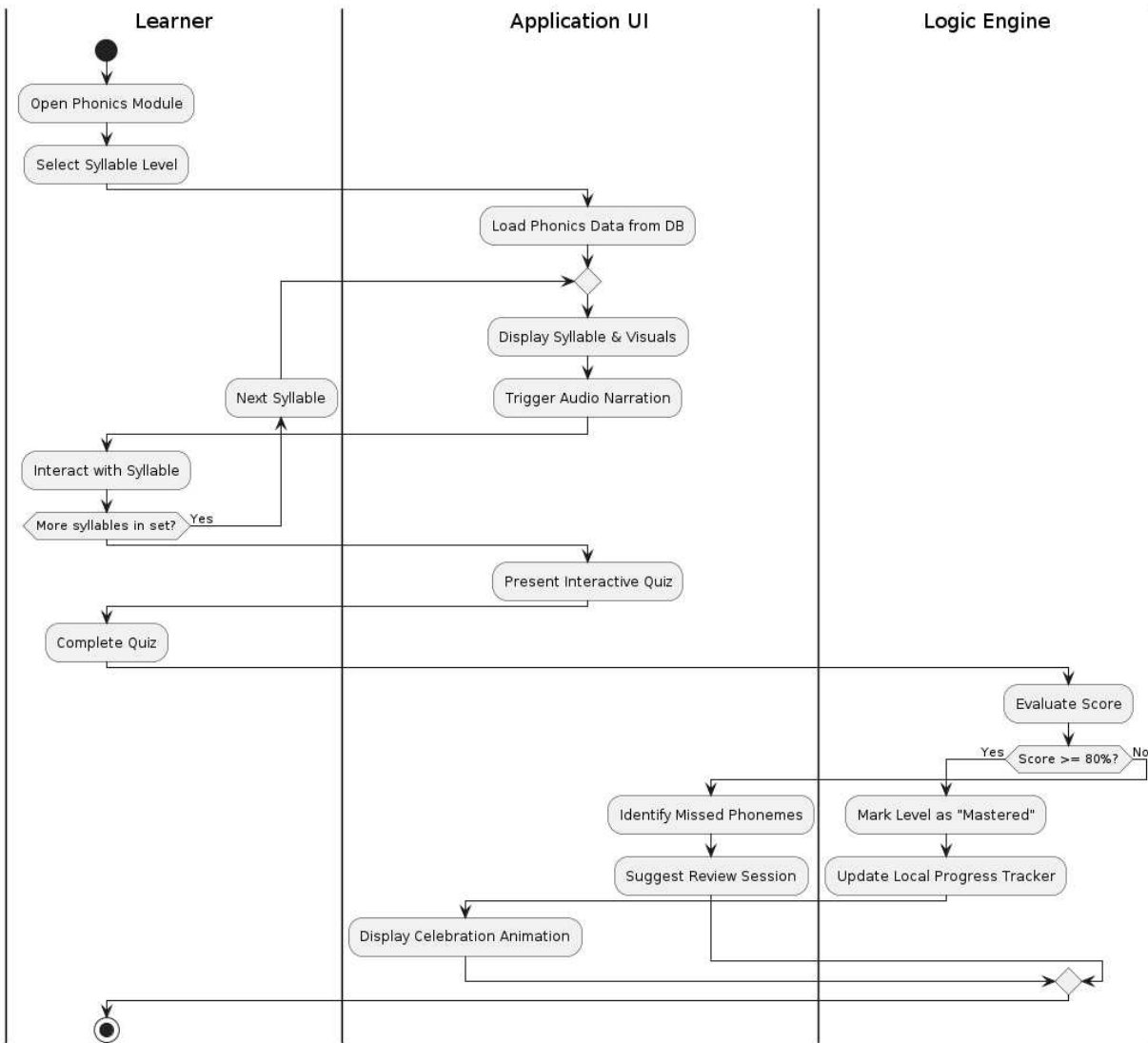


Figure 8 ACTIVITY DIAGRAM

STATE DIAGRAM

This State Machine Diagram diagram is vital for mapping the learner's journey. It ensures that the application logic correctly handles the flow from initial engagement to the mastery of a specific reading level, while maintaining strict rules for progression and data security.

1. Initial State (The Solid Circle): This represents the moment the application is launched.
2. Home/Main Menu State: The system starts here, allowing the learner to choose between the "Phonics" or "Stories" modules.
3. Active Learning State: Once a module is selected, the system enters a state where it displays content and plays audio. It stays in this state as long as the learner is interacting with the slides or pages.
4. Quiz/Assessment State: A transition occurs when the content ends, moving the system into an evaluation mode. Here, the system waits for user input to verify comprehension.
5. Evaluation State (The Decision Point): The system processes the score.
 - If the Guard Condition (e.g., Score > 80%) is met, the system transitions to the Mastered state.
 - If the score is low, it transitions back to the Active Learning state for remediation.
6. Parental Gateway State: A specific branch in the diagram shows the system entering a "Locked" or "Authorized" state where only a parent can access analytics or provide consent.
7. Final State (The Bullseye): This represents the user exiting the application or successfully completing the entire curriculum.

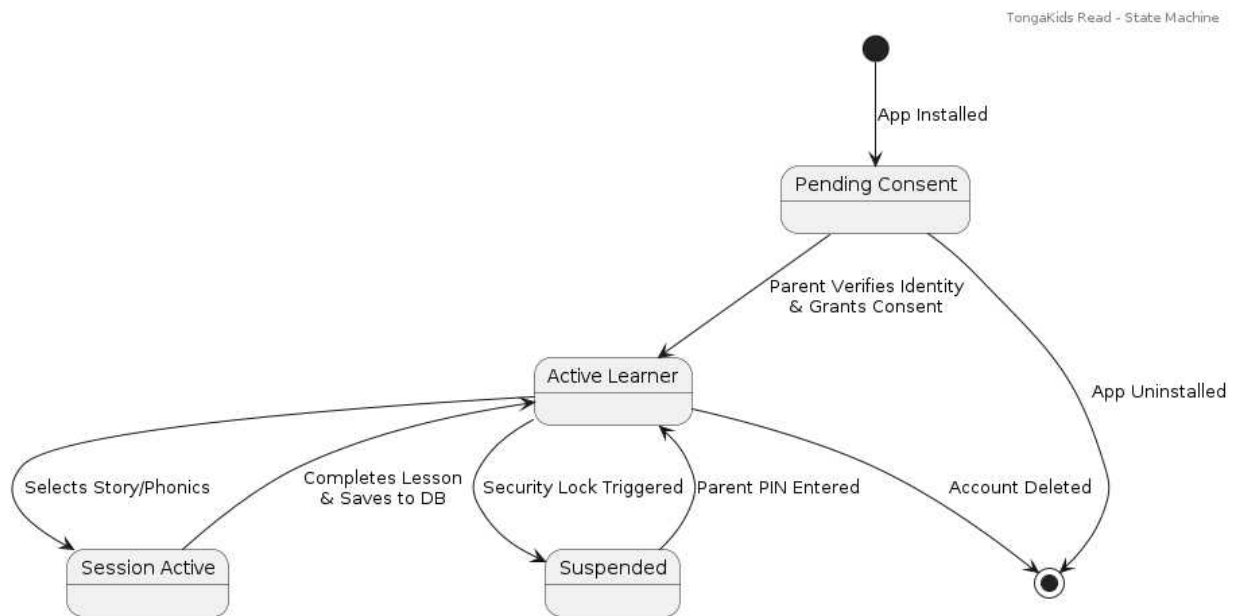


Figure 9STATE DIAGRAM

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